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ChatGrid: Intelligent Knowledge Q&A for Power Dispatching Control Based on Large Language Models and Retrieval-augmented Generation

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***Abstract*—As power systems become increasingly complex and intelligent, traditional power dispatching control systems struggle to effectively meet the demands of new digital and intelligent power grids. The issues of low digitalization, automation, and intelligence levels are becoming more pronounced. In response to these challenges, we propose an intelligent knowledge Q&A method for power dispatching control systems, named ChatGrid. This method leverages the robust language understanding and generation capabilities of Large Language Models (LLMs) and the comprehensive knowledge of the power dispatching control domain knowledge base, facilitated by Retrieval-augmented Generation (RAG). The process begins by vectorizing power dispatching control domain text to create a knowledge base, followed by preprocessing user questions and using similarity metrics for relevant vector retrieval. Subsequently, LLMs are utilized for command execution and answer generation, integrating domain knowledge and prompts. Experimental results indicate that our proposed method effectively enhances the intelligence level of knowledge Q&A in the power dispatching control domain and improves the context understanding, professionalism, interpretability, and accuracy of the answers provided by the LLMs. The proposed method has wide application prospects in the power dispatching control domain, offering digital, intelligent, and automated solutions for Q&A and command parts. Furthermore, this method can be extended to actual dispatching processes, providing dispatchers with key information retrieval and decision-making support.**

***Index Terms*—Large Language Models, Power Dispatching Control, Retrieval-augmented Generation, Vector Knowledge Base.**

I. Introduction

Knowledge Q&A in the control field is of great importance. In the frontier direction of the power dispatching control field, the smart encyclopedia of grid operation requires intelligent collection and recognition of professional knowledge and operation data in the control field. It aims to establish a professional knowledge base in the control field, explore the use of artificial intelligence LLM technology, construct a dispatch knowledge large model, and realize dispatch knowledge Q&A and data query functions. The goal is to train a "smart mentor" that can answer hundreds of questions without fail. On the other hand, the direction of intelligent auxiliary decision-making in dispatch operation requires the conversion of grid stability procedures, protection regulations, and other dispatch business procedures into electronic rule libraries. In cases of grid faults and abnormal changes in load flow, it matches the actual operating conditions of the grid, automatically pushes operating rule requirements, and intelligently generates auxiliary decisions for dispatch operation. Both are closely related to knowledge Q&A in the power dispatching control field.

However, the current knowledge Q&A in the control field faces many challenges. Firstly, the dialogue lacks interpretability: Existing knowledge Q&A systems in the control field often lack interpretability when answering questions. The responses to complex questions often appear illogical, inarticulate, overly rigid, and difficult to explain. Secondly, there is a lack of semantic understanding and professional knowledge: Existing knowledge Q&A systems in the control field lack semantic understanding capabilities, unable to accurately grasp the context of the question, and can only perform Q&A based on keyword matching and rules. They also lack professional knowledge in the field of power dispatching control, making it difficult to integrate and internalize professional knowledge. Thirdly, there is a lack of memory mechanism and context understanding ability: Most existing knowledge Q&A systems in the control field lack the necessary memory mechanism and context understanding ability. They cannot effectively use context information in multi-turn dialogues, which severely restricts their application in continuous interaction scenarios. Fourthly, the system scalability is low: The scalability of existing knowledge Q&A systems in the control field is low. They usually adopt a tightly coupled architectural design, and there are complex dependencies between various functional modules. This not only increases the development and maintenance costs of the system but also greatly limits its expansion capabilities. Fifthly, there is insufficient correlation between Q&A and operations: In the process of knowledge Q&A in the control field, knowledge Q&A and the use of tools for simulation operations are deeply related. Current systems cannot organically combine them, resulting in higher development costs and lower levels of intelligence. To address the aforementioned issues, this paper proposes ChatGrid, which is an intelligent knowledge Q&A for power dispatching control based on LLMs [4] and RAG [1].

II. Preliminaries and Related Works

A. Large Language Models

LLMs [5] are a type of natural language processing technology based on deep learning. They are extensively used in understanding and generating natural language text.These models, built on the Transformer architecture, possess a large number of parameters, enabling them to handle large-scale datasets.Through pre-training on extensive text data, LLMs [6] acquire complex patterns and structures of language. As a result, they excel in various language tasks, including text generation, translation, question answering, summarization, sentiment analysis, and more.

1) Advantages of LLMs: LLMs have several advantages [7], including: (1) Massive knowledge storage: LLMs can store and retrieve a vast amount of information, making them a valuable resource for knowledge-based tasks. (2) Strong semantic understanding: LLMs can understand the meaning of text at a deep level, enabling them to handle tasks that require a nuanced understanding of language. (3) Flexible knowledge combination and reasoning: LLMs can combine and reason with knowledge in flexible ways, allowing them to generate responses or make predictions that require integrating information from different sources. (4) Multi-turn interaction dialogue capability: LLMs can maintain context over multiple turns of dialogue, making them suitable for conversational applications. (5) Rapid adaptation to zero-shot/few-shot learning problems: LLMs can quickly adapt to new tasks with little or no training data, demonstrating their versatility and robustness.

2) Fine-tuning of LLMs: The fine-tuning phase refers to the process of adjusting the parameters of an LLM specifically designed for a particular task. The fine-tuning dataset includes strictly formatted instructions and related data, which may contain elements such as tables and images. This is done so that the LLM can learn how to handle and respond to information related to the specific task.

In more detail, fine-tuning of LLMs includes the following aspects. (1) Task-specific fine-tuning: This is a crucial step in the application of LLMs. After the initial pre-training phase, the model is further trained on a smaller, task-specific dataset. This process allows the model to adapt its previously learned language patterns to the specific requirements of the new task. (2) Data for fine-tuning: The data used for fine-tuning is carefully curated and formatted. It includes not only text but also other types of data relevant to the task, such as tables and images. This diverse data helps the model learn how to process and respond to a wide range of inputs. (3) Outcome of fine-tuning: After fine-tuning, the LLM is better equipped to handle the specific task. It can understand and generate responses that are more relevant and accurate, thereby improving its overall performance on the task.

3) Prompt Engineering: Prompt engineering is a method specifically optimized for LLMs. Its goal is to guide the LLM to generate more accurate and targeted output text by designing and adjusting the input prompts.

4) Tool Invocation in LLMs: LLMs possess the ability to invoke tools (functions). After predefining some tools (functions), when a user inputs a question (or instruction) to the LLM, the model determines whether a tool needs to be invoked based on text analysis. If necessary, it generates a structure for tool invocation.

The role of LLMs in tool invocation is twofold: firstly, it determines whether a tool needs to be invoked based on semantics; secondly, it extracts the function values required for tool invocation from the question or instruction.

5) Intelligent Chain: The intelligent chain is an approach to building applications that leverages the invocation capabilities of LLMs. It sequentially connects different tool modules involved in the intelligent processing of domain knowledge Q&A text. This paper provides benefits in terms of standardization, normalization, and modularization.

6) Memory Mechanism: The memory mechanism consists of memory modules and memory components, with each memory module being composed of several memory components. Memory components are essential supporting units that directly interface with the chain workflow and intelligent components. They are primarily responsible for recording significant information from the components within the chain workflow and supplying information for these components to access.

B. Retrieval-augmented Generation

LLMs, while powerful, do have their constraints. They demand significant computational resources and top-tier training datasets. Additionally, they necessitate regular retraining to keep their knowledge up-to-date, which makes them less ideal for quickly iterating new scenarios or assimilating vast amounts of new knowledge. The RAG technique [2] presents a viable approach to addressing these issues.

1) Concept Definition of RAG Method: The RAG approach [3] is a technique in natural language processing that merges the processes of information retrieval and text generation. Before the language model produces a response, RAG first extracts pertinent information from an extensive database of documents. This information is then utilized to steer the text generation process.

2) Process Overview of RAG Method: In the RAG approach [11], the process begins with the use of a pre-trained retrieval component to extract pertinent context information or documents from a knowledge repository. Subsequently, the information retrieved is supplied to a generation model. This model integrates the information into the text it generates, thereby enhancing the accuracy of responses, producing explanations, or improving the quality of the language model's output.

3) The Classic Paradigm of RAG: The classic paradigm of RAG [12], Naive RAG, is the most straightforward and classic form of RAG, which mainly includes four basic steps. (1) Extraction (Indexing): The document database is segmented into shorter pieces, known as chunks, and a vector index is constructed using an encoder. (2) Retrieval: Based on the similarity between the question and the chunks, relevant sections of documents are retrieved. (3) Prompt: A prompt for the LLM is created, which integrates the results of the query and the query statement itself. (4) Generation: The response to the question is generated, taking into account the context that has been retrieved.

4) Comparative Strengths and Weaknesses of RAG and Fine-tuning Techniques: The strengths and weaknesses of the RAG technique and the LLM fine-tuning technique can be compared as follows. The RAG technique has several advantages: it has the ability to directly update the knowledge base it retrieves from, allowing it to adapt to dynamic data environments. It is adept at leveraging external resources and is especially suitable for accessing documents or other structured or unstructured databases. It requires less data processing and fewer computational resources. It also offers relatively high interpretability, as the source of the answers can be traced back to specific data sources.The RAG technique has certain drawbacks: it might not allow for complete customization of the model's behavior or writing style. It involves the retrieval of information and the integration of external knowledge, which could result in increased latency. The strengths of the fine-tuning technique include: it has the ability to enhance the model's performance and efficiency by strengthening the base model's knowledge, modifying the output, and instructing complex commands, which allows the model to better mimic specific structures, styles, or formats. It can be utilized to tailor the model to cater to the requirements of specific fields or tasks, such as altering the model's behavior, writing style, or domain-specific knowledge. The fine-tuning technique has certain drawbacks: it demands substantial computational resources and high-quality training datasets. It might necessitate regular retraining to keep its knowledge up-to-date, which makes it less ideal for quickly iterating new scenarios or assimilating vast amounts of new knowledge.

III. Methodology

A. Enabling Knowledge Q&A for Power Dispatching Control with LLM and RAG

1) Enabling Knowledge Q&A for Power Dispatching Control with LLM: Utilizing the robust automatic Q&A capabilities of LLMs, we can realize knowledge Q&A that understands context and possesses extensive professional knowledge. We can accomplish fully automated Q&A in the power dispatching control domain. We can enhance the interpretability of Q&A in the power dispatching control domain. Specifically, we design an intelligent Q&A module for the power dispatching control domain and a tool invocation module, both based on LLMs. By integrating power dispatching control domain knowledge and preset prompts, we can generate flexible and precise answers, thereby improving the user's Q&A experience; we can also invoke tools as per user requirements.

2) Enabling Knowledge Q&A for Power Dispatching Control with RAG: By building a vector knowledge base for Q&A in the power dispatching control domain, we have integrated domain knowledge with the LLM, enabling precise comprehension and response to queries within the power dispatching control domain knowledge Q&A context. The vector knowledge base technology addresses two critical challenges: matching power dispatching control domain knowledge and expanding the scope of knowledge. This enhances the efficiency and accuracy of matching power dispatching control domain knowledge and allows for adaptation to the ever-evolving domain knowledge and power grid conditions.

B. ChatGrid: an Intelligent Knowledge Q&A Framework for Power Dispatching Control

In the foundational layer of the ChatGrid framework, we've amassed a wealth of documentation from the power dispatch control sector, including dispatch rules, dispatch procedure documents, and data on plants and equipment, among others. We've also utilized open-source LLMs like ChatGLM, DeepSeek, and Yi. Leveraging these resources, we've constructed a vector knowledge base for the dispatch control domain and a large-scale dispatch control model at the L1 layer. Building on this, we've employed LLM and RAG technologies to establish the application layer, and on top of this, we've developed an intelligent Q&A application. The framework of ChatGrid is as shown in Figure 1.

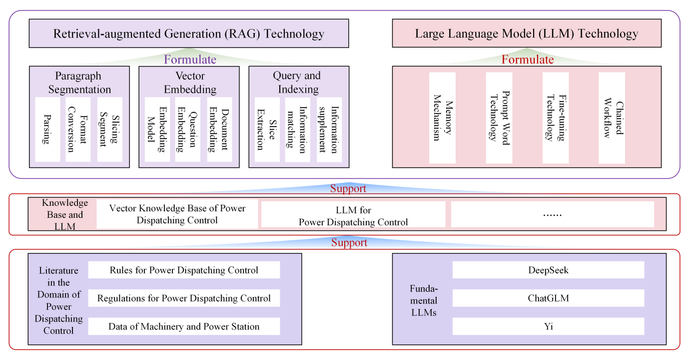

Fig. 1. The Framework of ChatGrid

In detail, the ChatGrid framework comprises three stages in automated Q&A, as follows. Text materials from the power dispatching control domain are fed into the knowledge base to construct a intelligent chain, thereby building a vector knowledge base for the power dispatching control domain. The ChatGrid system accepts user input text and stores it in the memory component, creating a comprehensive collection of user question text. The system, based on the user's question text, discerns whether it is a question or a command, and feeds it into the Q&A intelligent chain and the tool invocation intelligent chain, respectively. Below is a detailed explanation of the intelligent chain for knowledge base construction, the intelligent chain for intelligent Q&A, and the intelligent chain for tool invocation. The framework of ChatGrid is as shown in Figure 2.

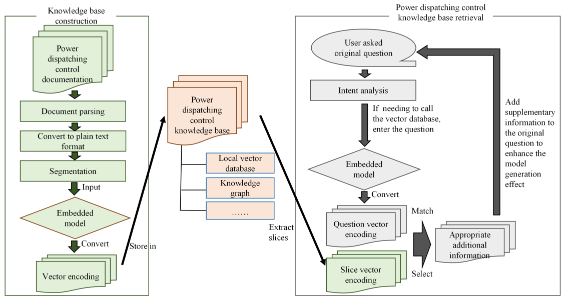

Fig. 2. The Process Diagram of ChatGrid

C. The Intelligent Chain for Knowledge Base Construction

Initially, we preprocess the text materials from the dispatcher training simulation domain by parsing and converting documents, transforming them into a plain text format and segmenting them. This process is formulated as:

$$T_p = f_s(f_t(f_p(T_o))) \tag{1}$$

in which T_o is the original text of power dispatching control, T_p is the text after preprocessing, f_p is the parsing function, f_t is the transforming function, and f_s is the segmenting function. Subsequently, we use an embedding model such as BERT [8], [9], [10] to generate vector encodings of the text, which we then store in the vector knowledge base for future retrieval. This process is formulated as:

$$T_e = f_e(T_p) \tag{2}$$

in which T_p is the text after the preprocessing, T_e is the text after embedding, and f_e is the embedding function.

D. The Intelligent Chain for Intelligent Q&A

The intelligent chain for intelligent Q&A is utilized for augmented Q&A generation. Initially, when a user inputs a question, we process the user's text using vector embedding and search for the most similar vector in the vector knowledge base using metrics like L2 similarity and Jaccard similarity. For user's text using vector X and knowledge base vector Y, this process is formulated as:

$$S_L(X,Y) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2} \tag{3}$$

in which x_i and y_i are the i-th components of X and Y, respectively. For two pieces of text X and Y, the formula for Jaccard similarity used in this paper is as follows:

$$S_J(A,B) = \frac{|A \cap B|}{|A \cup B|} \tag{4}$$

in which A and B are the word sets of text X and Y, respectively. $|A \cap B|$ is the size of the intersection of the two sets, and $|A \cup B|$ is the size of the union of the two sets. Based on the above L2 similarity and Jaccard similarity, this paper provides a comprehensive similarity calculation method, as shown below:

$$S = \lambda_L S_L + \lambda_J S_J \tag{5}$$

in which λ_L and λ_J are the weights of S_L and S_J, respectively. λ_L and λ_J can be adjusted according to the differences in power dispatching control text types.

Subsequently, we combine the text associated with the vector, preset prompts, and memories retrieved from the memory module. The LLM then generates an output that is returned to the user. This process is formulated as:

$$T_{out} = f_{LLM}(T_{in}, T_{Similar}, Prompt, Memory) \tag{6}$$

in which f_{LLM} is the LLM, T_{in} is the input text by users, T_{out} is the output text by the LLM, $T_{Similar}$ is the text corresponding to the most similar vector in the vector base, $Prompt$ is the preset prompts, and $Memory$ is the memories retrieved from the memory module.

E. The Intelligent Chain for Tool Invocation

Within the intelligent chain for tool invocation, we employ LLMs to facilitate command execution. Initially, we input the user's command text into the entity extraction component. The entities extracted are then fed into the integrity check component, where, in conjunction with prompt words, the LLM determines whether the elements required for tool invocation are complete. This process is formulated as:

$$E_{List} = f_{LLM}(T_{in}) \tag{7}$$

in which E_{List} is the list of entities extracted by the LLM, f_{LLM} is the LLM. If they are complete, the extracted entities are further input into the tool invocation component to call the tool; if they are incomplete, the extracted entities are fed into the prompt component to notify the user of the incomplete text.

IV. Experiments

To verify the effectiveness of the method proposed in this paper, we conduct a series of experiments. The experiments are mainly divided into two parts: comparison of accuracy and recall of knowledge base retrieval based on RAG and traditional method, and ablation experiments of similarity measurement in RAG retrieval.

A. Experiment Settings

First, we establish a knowledge base in the field of power dispatching control using dispatch regulations, procedural documents, equipment and plant data. Then, we conduct a search in this knowledge base and test its accuracy and recall. In addition, we conduct ablation experiments on the L2 similarity measurement used in the search to test the retrieval effect.

B. Comparison of Accuracy and Recall of Knowledge Base Retrieval Based on RAG and Traditional Method

We've developed a dataset for power dispatching control Q&A, comprising more than 5,000 lines of Q&A text from the power dispatch control domain. After embedding, we store it in the knowledge base for the power dispatch control domain and evaluate the retrieval accuracy and recall of both the RAG and traditional keyword matching methods. We apply 563 question text of users, which include positive and negative samples, to evaluate the accuracy and recall. The experimental outcomes indicate that the RAG method outperforms the traditional method in both accuracy and recall.

TABLE I
COMPARISON OF ACCURACY AND RECALL OF KNOWLEDGE BASE RETRIEVAL BASED ON RAG AND TRADITIONAL METHOD

Method	TP	FP	TN	FN	Accuracy	Recall
RAG	223	38	299	3	**92.72%**	**98.67%**
Keyword Match	15	0	337	211	62.52%	6.64%

C. Ablation Experiment of Similarity Measurement in RAG Retrieval

We carry out ablation experiments on similarity measurement using metrics such as L2 distance and Cosine distance. The results of these experiments indicate that the best performance is achieved when using L2 distance.

TABLE II
ABLATION EXPERIMENT OF SIMILARITY MEASUREMENT IN RAG RETRIEVAL

Similarity Measurement	Retrieval Accuracy
L2	**76.50%**
Cosine	24.40%
L2 and Cosine Hybrid	57.20%

V. CONCLUSION

In this paper, we introduce a robust LLM, a vector knowledge base for the control domain, RAG technology, a memory module, a tool invocation module, and a intelligent chain. These components address several significant challenges in the control domain's knowledge Q&A. Our experiments demonstrate that the LLM and RAG method we employ are effective for knowledge Q&A in the control domain. Furthermore, the authors plan to leverage the LLM, in conjunction with RAG technology, prompt word engineering, and command fine-tuning technology, to create a comprehensive and ”unstoppable” smart encyclopedia for power grid operations. This will facilitate highly digitized intelligent decision support for dispatch operations, assisting dispatchers in power dispatching control or even directly participating in it.

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